

Statistical & Numerical Methods Final Assignment

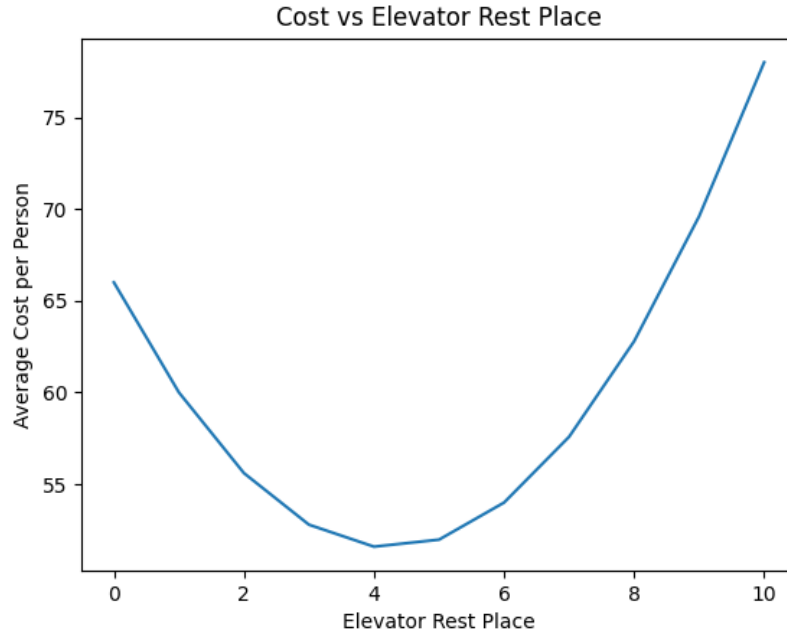
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1 Lift Waiting Location

The code solving this, with documentation, is attached as q1.py, the result it produces is floor 4.



1.1 Bonus Calculation

To find the optimal resting floor we'll start by analyzing the paths it takes, focusing on a single person. Through all the assumptions we can simplify the model to:

1. Person enters on floor 0.
2. Person go to floor X.
3. Person then goes to 3 random floors.
4. From the final floor he reaches, the person goes down to 0 and leaves.

in each movement step the elevator performs

1. Elevator goes to origin floor from its resting point.
2. Elevator goes to destination floor from the origin floor.
3. Elevator goes to the resting point from the destination floor.

Each ride's cost calculation is

$$C = |rest - origin| + |dest - origin| + |rest - dest| \quad (1)$$

we can notice that the middle element doesn't depend on the rest point so we can simply ignore it. Therefore the goal we want to minimize is

$$C = E(|rest - origin|) + E(|rest - dest|) \quad (2)$$

but from the symmetry of a sufficiently large random selection of people and trips we can simplify this to

$$C = 2E(|rest - X|) \quad (3)$$

where X is a floor. To this we add the trips as the mapping

$$\{0, f_1, f_2, f_3, f_4\} \rightarrow \{f_1, f_2, f_3, f_4, 0\} \quad (4)$$

we notice that the floor 0 appears twice and because the person travels from/to there in two separate trips, so when working with the floor numbers, we'll need to weight trips to/from the floor 0 at double the weight of the others. We can now calculate

$$E(|rest - X|) = \frac{rest}{5} + \frac{4}{5} \cdot \frac{1}{10} \sum_{X=1}^{10} |rest - X| \quad (5)$$

we take the derivative to find the optimum

$$\frac{dE}{drest} = \frac{1}{5} + \frac{4}{50} \sum_{X=1}^{10} \text{sgn}(rest - X) \quad (6)$$

and

$$\frac{4}{50} \sum_{X=1}^{10} \text{sgn}(rest - X) + \frac{1}{5} = 0 \quad (7)$$

Which is a pain to solve so we turn to tricks. If we assume there are k floors below the resting point, this means the expression inside the sum will be $+1$ k times, and -1 $10 - k - 1$ times. Therefore

$$D = \frac{4}{50} \sum_{X=1}^{10} \text{sgn}(rest - X) + \frac{1}{5} = \frac{1}{5} + \frac{4}{50} (k + k - 9) = \frac{10 - 36 + 8k}{50} = \frac{8k - 26}{50} \quad (8)$$

plugging in the different possible values for k

$$D(0) = -\frac{26}{50} \quad D(1) = -\frac{18}{50} \quad D(2) = -\frac{10}{50} \quad D(3) = -\frac{2}{50} \quad D(4) = \frac{6}{50} \quad (9)$$

so our two suspect floors are 3 and 4. We look at the values at the floors:

$$E(|3 - X|) = \frac{3}{5} + \frac{4}{50} \sum_{X=1}^{10} |3 - X| = \text{calculator} = \frac{77}{25} \quad (10)$$

$$E(|4 - X|) = \frac{4}{5} + \frac{4}{50} \sum_{X=1}^{10} |4 - X| = \text{calculator} = \frac{74}{25} \quad (11)$$

so floor 4 gives a lower expected value, meaning it is our optimal floor.

2 Peaches and Oranges

Since we have no information on any distributions or chances, we'll treat the fruit counts as being distributed poisson.

2.1 Oranges in the 2376 day

We can use the total number of peaches counted to find the expected value of peaches per day

$$\lambda = 10500/7 = 1500 \quad (12)$$

so the number of oranges is

$$N = 2376 - \lambda = 876 \quad (13)$$

The number of fruit counted the specific day is exact so has no variance. The number of peaches has variance which comes both from the variance on the actual count of peaches and from the daily count fluctuating around the average we calculate.

$$\sigma_{peach, week} = \sqrt{10500} \sim 102.5 \quad (14)$$

but since we divided it by 7 to look at each day

$$\sigma_{peach, day} = \frac{102.5}{7} \sim 14.6 \quad (15)$$

then the variance due to the actual daily number not being the exact average

$$\sigma_{day} = \sqrt{1500} \sim 38.7 \quad (16)$$

so the number of oranges that day is

$$N = 876 \pm 14.6 \pm 38.7 \quad (17)$$

and since those are independent the total error is

$$\sigma_{total} = \sqrt{(14.6)^2 + (38.7)^2} \sim 41.4 \quad (18)$$

so

$$N = 876 \pm 41.4 \quad (19)$$

2.2 Expected Daily Count

This time the count of total fruit is also uncertain. We treat it as a single measurement from a poisson distribution meaning

$$\lambda_{fruit} = 2376, \quad \sigma_{fruit} \sim 48.7 \quad (20)$$

and as for the count of peaches, this time we don't have the second source of uncertainty, since we don't need to consider whether the specific day was average or not, since we're looking at the average itself, so

$$\lambda_{peaches} = 1500, \quad \sigma_{peaches} = 14.6 \quad (21)$$

which gives us

$$N = 876 \pm 48.7 \pm 14.6 \tag{22}$$

and again summing independent errors

$$\sigma = \sqrt{48.7^2 + 14.6^2} \sim 50.8 \tag{23}$$

so

$$N = 876 \pm 50.8 \tag{24}$$

3 Resistors

Since the resistors are connected in parallel, we'll find the equivalent total resistance using

$$\frac{1}{R_{tot}} = \sum \frac{1}{R} \Rightarrow R_{tot} = \frac{R_1 R_2 R_3}{R_1 R_2 + R_1 R_3 + R_2 R_3} \quad (25)$$

giving us

$$R_{tot} = 3.3\text{k}\Omega \quad (26)$$

3.1 Total error

The error on the total resistance will have to be calculated using error propagation

$$\sigma^2 = \left(\frac{\partial R_{tot}}{\partial R_1} \right)^2 \sigma_{R1}^2 + \left(\frac{\partial R_{tot}}{\partial R_2} \right)^2 \sigma_{R2}^2 + \left(\frac{\partial R_{tot}}{\partial R_3} \right)^2 \sigma_{R3}^2 \quad (27)$$

$$= 3 \left(\frac{R_2 R_3 (R_1 R_2 + R_1 R_3 + R_2 R_3) - R_1 R_2 R_3 (R_2 + R_3)}{(R_1 R_2 + R_1 R_3 + R_2 R_3)^2} \right)^2 \sigma_{R1}^2 \quad (28)$$

$$= 3 \left(\frac{R_1 R_2^2 R_3 + R_1 R_2 R_3^2 + R_2^2 R_3^2 - R_1 R_2^2 R_3 - R_1 R_2 R_3^2}{(R_1 R_2 + R_1 R_3 + R_2 R_3)^2} \right)^2 \sigma_{R1}^2 \quad (29)$$

$$= 3 \left(\frac{R^4}{9R^4} \right)^2 \sigma_R^2 = \frac{1}{27} \sigma_R^2 \quad (30)$$

$$= \frac{1}{27} (0.05 \cdot 10\text{k}\Omega)^2 \quad (31)$$

$$= 9259\Omega^2 \quad (32)$$

so

$$\sigma = 96\Omega \quad (33)$$

where we used the fact the resistors are identical several times.

3.2 A single resistor

This time we only have a single source of error.

$$\sigma = 0.05 \cdot 3.3\text{k}\Omega = 165\Omega \quad (34)$$

3.3 in series instead

To get the same voltage we need the same resistance. The total resistance we found was $3.3\text{k}\Omega$ so we need three resistors each $1.1\text{k}\Omega$. The equation for total resistance is

$$R_{tot} = R_1 + R_2 + R_3 \quad (35)$$

so

$$\sigma^2 = 3 \left(\frac{\partial R_{tot}}{\partial R} \right)^2 \sigma_R^2 = 3\sigma_R^2 = 3 \cdot (0.05 \cdot 1.1\text{k}\Omega)^2 = 9075\Omega^2 \quad (36)$$

so

$$\sigma = 95\Omega \tag{37}$$

3.4 Different resistances vs the same

Yes there is a difference. Since they are independent elements, we sum the squares of the standard deviations then take the square root. If we look at the limit where one resistor has all the resistance and the others have resistance 0, the error is the same as the error on the individual resistor. Moving resistance to another resistor takes the value out from the square in the error calculation and moves it to a multiplication inside the root, which reduces it.